

Rui SHENG

VisLab, advised by Prof. Huamin QU
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EDUCATION

- Ph.D. Computer Science and Engineering, the Hong Kong University of Science and Technology (HKUST), China, 2021-2026
- B.S. Computer Science and Technology (Rank: 2/334), Wuhan University (WHU), China, 2017-2021

RESEARCH INTERESTS

Human-AI Collaboration, Data Visualization, Biomedicine/Biochemistry Scientific Discovery, Clinical Applications, LLM-based Agentic Systems

AWARDS

- 2026 Redbird Scholarship in HKUST
- 2026 IEEE Computer Graphics and Applications Review Award
- 2025 EMNLP Oral Presentation (5%)
- 2025 IEEE VIS Best Paper Honorable Mention Award (5%)
- 2025 Redbird Scholarship in HKUST
- 2021 Hong Kong Ph.D. Fellowship Scheme (4%, HK\$331,200 annually)
- 2020 National Scholarship in Wuhan University (6/334, the highest scholarship in WHU)
- 2019 National Scholarship in Wuhan University (6/342, the highest scholarship in WHU)

PUBLICATIONS

First-author

- 2026 **Rui Sheng**, Furui Cheng, Chuhan Shi, Sobhan Lotfi, Shiyi Liu, Adam Perer, and Huamin Qu. "Design Patterns of Human-AI Interfaces in Healthcare." *IJHCS*.
- 2026 Haoyu Dong+, **Rui Sheng**+, Shuhao Zhang, Yushi Sun, Dingyang Wu, Hanxiang Chao, Olexandr Isayev, Huamin Qu, Yuyang Wu, and Yanna Lin. "MoleculeCanvas: Interactive In-Context Prompting for LLM-Assisted Molecular Optimization." *ACM UIST*.
- 2026 **Rui Sheng**+, Yukun Yang+, Chuhan Shi, Yanna Lin, Zixin Chen, Huamin Qu, and Furui Cheng. "DiLLS: Interactive Diagnosis of LLM-based Multi-agent Systems via Layered Summary of Agent Behaviors." *ACM CHI*.
- 2025 **Rui Sheng**, Zelin Zang, Jiachen Wang, Yan Luo, Zixin Chen, Yan Zhou, Shaolun Ruan, and Huamin Qu. "CellScout: Visual Analytics for Mining Biomarkers in Cell State Discovery" *IEEE TVCG*.

- 2025 **Rui Sheng**, Yifang Wang, Xingbo Wang, Shun Dai, Qingyu Guo, Tai-Quan Peng, Huamin Qu, and Dongyu Liu. “EMINDS: Understanding User Behavior Progression for Mental Health Exploration on Social Media.” *IEEE TVCG*.
- 2025 **Rui Sheng**, Xingbo Wang, Jiachen Wang, Xiaofu Jin, Zhonghua Sheng, Zhenxing Xu, Suraj Rajendran, Huamin Qu, and Fei Wang. “TrialCompass: Visual Analytics for Enhancing the Eligibility Criteria Design of Clinical Trials.” *IEEE VIS*.
- 2023 **Rui Sheng**, Leni Yang, Haotian Li, Yan Luo, Ziyang Xu, Zhilan Zhou, David Gotz, and Huamin Qu. “Knowledge Compass: A Question Answering System Guiding Students with Follow-Up Question Recommendations.” *ACM UIST Symposium*.
- **Rui Sheng**, Yonggang Li, Chuhan Shi, Huamin Qu, and Adam Perer. “AIvidence: Reconciling Patient-Specific AI Explanations with Evidence-Based Medicine for Clinical Diagnosis.” *Fast-track to TVCG*.
- **Rui Sheng**+, Yukun Yang+, Xinli Zhu, Yushi Sun, Zixin Chen, Haobo Li, Xiangzhe Kong, and Yuyang Wu. “What, Where, and How to Edit: Benchmarking Instruction Following for Localized Molecular Editing.” *Under review*.

Corresponding-author

- 2026 Chuhan Shi, Zijian Guo, Zelin Zang, Chengbo Zheng, Ding Ding, and **Rui Sheng***. “CellPrism: A Visual Analytics System for Exploring AI-Driven Virtual Cells in Drug Discovery.” *IEEE VIS (Best Paper Award, top 1%)*.
- 2025 Leixian Shen+, Yan Luo+, **Rui Sheng***, Yujia He, Haotian Li, Leni Yang, and Huamin Qu. “StoryLensEdu: Personalized Learning Report Generation through Narrative-Driven Multi-Agent Systems.” *IEEE PacificVis*.
- Chuhan Shi, Xiaoquan Ren, Yifang Wang, Junze Li, Yushi Sun, Yawen Luo, and **Rui Sheng***. “A Survey of Human-AI Collaboration for Scientific Discovery.” *Under review*.
- Shuo Yan, Xiaolin Wen, Shaolun Ruan, Yanjie Zhang, Jiaming Mi, Yushi Sun, Huamin Qu, and **Rui Sheng***. “InconLens: Interactive Visual Diagnosis of Behavioral Inconsistencies in LLM-based Agentic Systems.” *Under review*.
- Yanna Lin, Shaojie Xu, Wenshuo Zhang, Yushi Sun, Zixin Chen, Yanjie Zhang, and **Rui Sheng***. “A Survey of LLM-based Multi-agent Systems in Medicine.” *Under review*.
- Peng Kuai, Yukun Yang, Shaolun Ruan, Junchi Xu, Yanjie Zhang, Lin Zhang, Min Zhu, and **Rui Sheng***. “Within the MDT Room: Situated in Multidisciplinary Team–Grounded Agent Debate for Clinical Diagnosis.” *Under review*.
- Junchi Xu, Qipeng Wang, Yang Ouyang, Yushi Sun, Zelin Zang, Ziyi Deng, Yimeng Wang, Yinjie Zhou, **Rui Sheng***, and Min Zhu*. “CellDecipher: Visual Analytics for Supporting Human–LLM Collaboration in Cell Type Annotation.” *Under review*.
- Chuhan Shi, Xiaoquan Ren, Sicheng Song, Haobo Li, **Rui Sheng***, and Yushi Sun*. “Are LLMs Ready for Scientific Discovery? A Capability-Oriented Benchmark for AI Scientists.” *Under review*.
- Yushi Sun, Yanjie Zhang, and **Rui Sheng***. “The Personalization Mirage: How LLMs Fabricate User Profiles, and Why Self-Monitoring Misleads.” *Under review*.

Co-author

- 2026 Yanjie Zhang, Yafei Li, **Rui Sheng**, Zixin Chen, Yanna Lin, Huamin Qu, Lei Chen, and Yushi Sun. “Navigating the Mirage: A Dual-Path Agentic Framework for Robust Misleading Chart Question Answering.” *ACM MM*.
- 2026 Yurui Xiang, Xingyi Mao, **Rui Sheng**, Zixin Chen, Zelin Zang, Yuyang Wu, Haipeng Zeng, Huamin

- Qu, Yushi Sun, and Yanna Lin. “VeriLLMed: Interactive Visual Debugging of Medical Large Language Models with Knowledge Graphs.” *IEEE VIS*.
- 2025 Shaolun Ruan, **Rui Sheng**, Xiaolin Wen, Jiachen Wang, Tianyi Zhang, Yong Wang, Tim Dwyer, and Jiannan Li. “Qualitative Study for LLM-assisted Design Study Process: Strategies, Challenges, and Roles.” *IEEE VIS (Best Paper Honorable Mention Award)*.
- 2025 Qipeng Wang, Shaolun Ruan, **Rui Sheng**, Yong Wang, Min Zhu, and Huamin Qu. “TrajLens: Visual Analytics for Constructing Cell Development Trajectories in Cross-Sample Analysis.” *IEEE VIS*.
- 2025 Zixin Chen, Sicheng Song, Kashun Shum, Yanna Lin, **Rui Sheng**, and Huamin Qu. “Unmasking Deceptive Visuals: Benchmarking Multimodal Large Language Models on Misleading Chart Question Answering.” *ACM EMNLP (Oral)*.
- 2025 Xingbo Wang, Samantha L. Huey, **Rui Sheng**, Saurabh Mehta, and Fei Wang. “SciDaSynth: Interactive Structured Knowledge Extraction and Synthesis from Scientific Literature with Large Language Model.” *Campbell Systematic Reviews*.
- 2025 Jiachen Wang, Zikun Deng, Dazhen Deng, Xingbo Wang, **Rui Sheng**, Yi Cai, and Huamin Qu. “Empowering Multimodal Analysis with Visualization: A Survey.” *Computer Science Review*.
- 2025 Qipeng Wang, **Rui Sheng**, Shaolun Ruan, Xiaofu Jin, Chuhan Shi, and Min Zhu. “SynthLens: Visual Analytics for Facilitating Multi-step Synthetic Route Design.” *IEEE TVCG*.
- 2023 Junxiu Tang, **Rui Sheng**, Yifang Wang, Xinhuan Shu, Xiaojiao Chen, Tan Tang, and Yingcai Wu. “Posts with no Response: The Island of Loneliness.” *IEEE VIS Arts Program (accepted in two tracks - Pictorial and Artwork)*.
- 2021 Min Hu, Ruimin Hu, Xiaocheng Wang, and **Rui Sheng**. “Unsupervised Temporal Attention Summarization Model for User Created Videos.” *IEEE MMM*.
- Qipeng Wang, **Rui Sheng**, Yafei Li, Huamin Qu, Min Zhu, and Yushi Sun. “MedKGI: Iterative Differential Diagnosis with Medical Knowledge Graphs and Information-Guided Inquiring.” *Under review*.
- Jianben He, **Rui Sheng**, Xingbo Wang, Kam Kwai Wong, Xinhuan Shu, and Huamin Qu. “Engager: A Visual Analytics System for Multi-person and Multimodal Engagement Analysis in Online Teaching Videos.” *Under review*.
- Yifang Wang, **Rui Sheng**, Erzhuo Shao, Yifan Qian, Haotian Li, Nan Cao, and Dashun Wang. “Figures as Interfaces: Toward LLM-Native Artifacts for Scientific Discovery.” *Under review*.
- Shaolun Ruan, Zhikang Wang, Furui Cheng, **Rui Sheng**, Kai Qian, Jiangning Song, and Yong Wang. “ImaGene: A Multimodal Visualization Tool for Gene-powered Cancer Prognosis.” *Fast-track to CGF*.
- Chuhan Shi, Youyuan Zhang, Zijian Xu, **Rui Sheng**, Chengbo Zheng, Yanna Lin, and Yige Xu. “Ideation LLM: An Interactive Tool for Generating Interdisciplinary Research Ideas.” *Under review*.
- Ziyi Deng, **Rui Sheng**, Qipeng Wang, Dingdong Liu, Chuhan Shi, Junchi Xu, Yimeng Wang, Zixin Chen, and Min Zhu. “Simulemma: An Interactive Multi-agent Simulation System for Designing Ill-structured Interdisciplinary Group Assignments.” *Under review*.
- Kexin Han, Philipp Blankenhagen, Sicheng Song, **Rui Sheng**, and Huamin Qu. “Discuss Again: Turning One-off Offline Group Discussions into Repeatable Practice for Interaction Competence.” *Under review*.
- Chuhan Shi, Zhijie Chen, Yunfei Wang, Wenshuo Zhang, **Rui Sheng**, Dingdong Liu, and Zixin Chen. “CrossSense: Bridging Communication Gaps in Cross-Disciplinary Semi-Structured Interviews for Empathy Building.” *Under review*.
- Hanxiang Chao, Yihan Bai, **Rui Sheng**, Tianle Li, and Yushi Sun. “STALE: Can LLM Agents Know When Their Memories Are No Longer Valid?” *Under review*.

- Zixin Chen, Peng Liu, **Rui Sheng**, Haobo Li, Jianhong Tu, Xiaodong Deng, Kashun Shum, Dayiheng Liu, and Huamin Qu. “Are Agents Ready to Teach? A Multi-Stage Benchmark for Real-World Teaching Workflows.” *Under review*.
- Zixin Chen, Peng Liu, Haobo Li, **Rui Sheng**, Jianhong Tu, Xiaodong Deng, Fei Huang, Kashun Shum, Dayiheng Liu, and Huamin Qu. “WILDTRACE: Benchmarking Natural Evidence Trails in Long-Context Reasoning.” *Under review*.
- Wenshuo Zhang, Zijian Xu, Zisen Qin, **Rui Sheng**, Yushi Sun, Huamin Qu, and Chuhan Shi. “Fewer Clarifications, Better Code: Benchmarking Cross-Session Personalized Ambiguity Adaptation in Coding Assistants.” *Under review*.
- Dongjie Yang, Siyan Lin, Leixian Shen, **Rui Sheng**, Huamin Qu, and Zixin Chen. “TACT: Taxonomy-Aligned Post-Training for Pedagogically Adaptive English Tutoring.” *Under review*.
- Chuhan Shi, Xinli Zhu, Zelin Zang, **Rui Sheng**, and Zixin Chen. “EcoCell: Visual Analytics of Cell Differentiation within the Cellular Microenvironment.” *Under review*.
- Yonggang Li, Boqian Han, **Rui Sheng**, Yushi Sun, and Zixin Chen. “RegulaTutor: A Theory-Informed Framework for Long-Term Personalized Learning.” *Under review*.
- Xinli Zhu, Rongchang Li, **Rui Sheng**, Kezhao Chen, Zizhao Zhang, Chuhan Shi, and Yushi Sun. “Follow the Edit: Counterfactual Diagnostics and On-Policy Rank Distillation for Composed Image Retrieval.” *Under review*.
- Haobo Li, Wenshuo Zhang, Wenxiao Zhao, Eunseo Jung, **Rui Sheng**, Yushi Sun, Peiqin Zhuang, Hao Chen, and Fenghua Ling. “RISR: Residual-Informed Scientific Equation Discovery with Large Language Models.” *Under review*.

INTERNSHIP

THU	THUNLP Group (Prof. Yang Liu), Institute for AI Industry Research, Tsinghua University (THU), China, Onsite, 2026.
CMU	Data Interaction Group (Prof. Adam Perer), Human-Computer Interaction Institute, School of Computer Science, Carnegie Mellon University (CMU), the United States, Onsite, 2025.
NU	Center for Science of Science & Innovation (Prof. Dashun Wang), Kellogg School of Management, Northwestern University (NU), the United States, Remote, 2024.
Cornell	Wang Lab (Prof. Fei Wang), Weill Cornell Medicine Graduate School of Medical Sciences, Cornell University, the United States, Remote, 2023.
UC Davis	VIA Lab (Prof. Dongyu Liu), Department of Computer Science, College of Engineering, the University of California at Davis (UC Davis), the United States, Remote, 2023.

PROJECTS

2025-26	Project member. AI-assisted Virtual Reality English Speaking Program for Secondary Students. HK\$30M by the Quality Education Fund, Hong Kong
2025-26	Project member. From Classroom to Career: LLM-Driven Curriculum Design and Talent Development in the Age of AI and Emerging Technologies. HK\$600K by the Education University of Hong Kong & the Hong Kong University of Science and Technology Joint Centre for Artificial Intelligence Application
2024-27	Project leader. Visual Analytics of Student-AI Interaction Data for Generative AI-Assisted Learning. HK\$1M by the General Research Fund, the Research Grants Council of Hong Kong

- 2023-25 Project leader. AI-Powered, Personalized Analysis and Feedback Platform for K-12 Education in Computational Thinking Skills.
 HK\$450K by the ITF project & CUHK project
- 2021-23 Project member. AI-powered Audience Engagement Analysis for Virtual Communication.
 HK\$2M by the ITF project & industry project

INVITED TALKS

- 2026 Graphics And Mixed Environment Symposium, hosted by CAD&CG.
 Topic: Human-Centered Visual Analytics for Biomedical Scientific Discovery: From Data to Model.
- 2026 Microsoft Research Lab - Asia (Hong Kong).
 Topic: Human-Centered Visual Analytics for Biomedical Scientific Discovery: From Data to Model.
- 2025 Nanjing University, hosted by Prof. Ke XU.
 Topic: Human-AI Collaboration in Scientific Discovery.

SERVICE

Academic Peer Review

- The International Conference on Learning Representations (ICLR)
- Journal of Medical Internet Research (JMIR)
- The Meeting of the Association for Computational Linguistics (ACL)
- Eurographics Conference on Visualization (EuroVis)
- The Conference on Human Factors in Computing Systems (CHI & CHI-LBW)
- Visual Informatics (VI)
- The Conference on Computer-Supported Cooperative Work and Social Computing (CSCW).
- The Pacific Visualization (PacificVis - TVCG track & Conference track).
- The China Visualization and Visual Analytics Conference (ChinaVis).
- The Conference on Information and Knowledge Management (CIKM).

Teaching Assistant

- 2024 HKUST COMP 2211 - Exploring Artificial Intelligence
- 2024 HKUST COMP 1021 - Introduction to Computer Science
- 2023 HKUST COMP 4321 - Search Engines for Web and Enterprise Data

Lab Service

- 2025-26 In charge of the management of the VisLab
- 2024-25 In charge of the BBS of the VisLab
- 2022-24 In charge of the publicity of our group (including WeChat, Zhihu, RedBook, and Twitter)
- 2021-22 In charge of maintaining the VisLab homepage